# Smart Water Logging, Flood Prevention and Post Flood Event Management System

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Project Proposal Report

Jayanath H.P.K.N – IT22170002

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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## DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

| Name | Student ID | Signature |
|---|---|---|
| Jayanath H.P.K.N | **IT22170002** |  |

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

| Name | Date | Signature |
|---|---|---|
| Deemantha Siriwardana | 14/03/2026 |  |

# ABSTRACT

Flooding is one of the most common natural disasters that affect many developing countries around the globe, including Sri Lanka. Floods are caused mainly by heavy monsoon rains, overflowing rivers, and poor drainage systems. Floods are one of the most devastating natural disasters that affect many countries worldwide. After a flood disaster, the risk of waterborne and vector-borne infectious diseases such as Leptospirosis, Dengue fever, Diarrhoea, Skin infections, and Respiratory infections is increased. In many disaster situations, healthcare workers are forced to rely mainly on manual reporting and physical medical examinations to detect people who are at risk of disease. The traditional methods are time-consuming and often result in delays in the detection of early symptoms of disease. In this regard, this research proposes a concept for developing an AI-based "Post Flood Disease Management Module" as part of a smart disaster response system. The proposed system will help in early detection of flood-related health risks by incorporating a system of health reporting based on symptoms, along with facial scan analysis. Individuals who are affected by floods can submit their health-related symptoms via a web or mobile interface, while facial scan images will be used for analyzing health-related factors such as fatigue, swelling, discoloration, and redness of eyes, which are considered health indicators for individuals affected by floods. The system is able to generate automated health alerts and recommendations, which are then displayed on a centralized dashboard that can be utilized by various healthcare response teams as well as disaster management agencies. This helps medical response teams to quickly identify people who are at greater risk in disaster response operations. The expected outcome of the research is the development of an intelligent health monitoring module that will help in the early detection of diseases, the management of healthcare after a flood disaster, as well as quick decision-making during disaster response operations. The system will also help in the management of disaster response operations due to the integration of healthcare monitoring with smart disaster response systems.

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### LIST OF ABBREVIATIONS

| Abbreviation | Description |
|---|---|
| DMC | Disaster Management Center |
| AI | Artificial Intelligence |
| SVM | Support Vector Machine |
| CNN | Convolutional Neural Networks |
| SDG | Sustainable Development Goals |
| DSR | Design Science Research |
| NGO | Non-Government Organization |
| SaaS | Software as a Service |

*Table 1 - List of Abbreviation*

# 1. INTRODUCTION

## 1.1 Background and Context

One of the most frequent natural disasters in the world, flooding has a major impact on developing nations with tropical climates. Heavy monsoon rains, overflowing rivers, and poor drainage infrastructure are the main causes of Sri Lanka's frequent severe floods. In addition to causing infrastructure and property damage, these floods pose a major threat to the public's health in the impacted areas.

Waterborne and vector-borne illnesses are more likely to spread after floods due to contaminated water, unsanitary conditions, and overcrowded shelters. Leptospirosis, dengue fever, diarrhoea, skin infections, and respiratory ailments are common post-flood illnesses. To stop widespread disease outbreaks, early detection of these health risks is crucial.

Every year, floods in Sri Lanka force thousands of people to flee their homes. Over 200,000 people were impacted by floods during the 2024 monsoon season, according to the DMC, with severe flooding occurring in several districts, including Ratnapura, Kalutara, Galle, and Matara.

One of the worst natural disasters to hit the nation in recent decades occurred in 2025 due to severe flooding and landslides brought on by Cyclone Ditwah and heavy monsoon rainfall. Over 2.2 million people were impacted in 25 districts, according to reports, and at the height of the emergency, over 300,000 people were displaced. Due to numerous homes being destroyed or situated in high-risk areas, over 153,000 people were still displaced as of early 2026. Hundreds of people died as a result of the disaster, which also seriously disrupted relief and healthcare services and caused extensive damage to homes and infrastructure.

Healthcare professionals have a difficult time identifying people who are at a high risk of contracting disease related to flooding in such circumstances.

At the moment, manual reporting and in-person medical examinations are the primary methods used for post-flood health monitoring. These techniques take a lot of time and frequently cause delays in the diagnosis of illness symptoms. Because o this, outbreaks might spread before medical personnel can stop them.

The suggested system adds a Post-Flood Disease Management Module that facilitates early health risk identification in order to address this problem. Fatigue, skin discoloration, and swelling are examples of visible health indicators that the system detects by combining facial scan analysis with symptom-based inputs from afflicted individuals via a mobile or web interface. To estimate disease risk levels, rule-based machine learning models are applied to the collected data.

In order to help medical professionals prioritize medical support for high-risk individuals, the system then generates health alerts and recommendations for medical teams and displays the results on a central dashboard.

In particular, this solution in line with the SDGs of United Nations.

- SDG 3 – Well-being and Good Health
- SDG 11 – Sustainable Cities and Communities

- SDG 13 – Climate Action

Additionally, because the study combines IOT systems, artificial intelligence, and data-driven decision support technologies to enhance disaster response, it falls under the Software Systems and Technologies (SST) research cluster.

## 1.2 Literature Review

The use of technology to enhance disease detection and disaster management following natural disasters has been investigated by a number of researchers.

CNNs can be used to analyze visual features, as demonstrated by Mohanty et al. (2020), who used image- based analysis to show how deep learning techniques are effective in identifying plant diseases. In the medical field, similar image processing methods can be used to recognize visible symptoms in people's faces.

Esteva et al. (2019) demonstrated in another study that deep learning models can identify skin disease using medical images with dermatologist-level accuracy. Their research demonstrates how image analysis can be used to automatically identify medical conditions.

Khan et al. (2021) suggested a mobile-based health monitoring system for populations affected by disasters as part of disaster management research. The vicitims health information was gathered by the system and sent to emergency response teams. The system was helpful, but it lacked automated visual health analysis and mostly relied on manual symptom reporting.

In a similar vein, Rahman et al. (2022) created an Internet of Things-based disaster management system that incorporates communication networks and sensors to aid in emergency response. However, rather than post-disaster healthcare, their system was mainly concerned with infrastructure monitoring.

Another study by Li et al. (2023) investigated the application of AI in emergency public health monitoring. In order to identify possible disease outbreaks, their system examined symptom data from mobile applications. Nevertheless, the strategy was not integrated with real-time disaster management systems.

In order to monitor disease outbreaks, current products like HealthMap and mobile-based epidemic surveillance systems also gather health data from communities. Although these systems are helpful for large-scale health monitoring, they lack facial analysis and direct integration with rescue management systems, and they were not created with post-flood disaster environment in mind.

The potential of AI- based image recognition and mobile health monitoring systems is generally demonstrated by prior research. However, the majority of current solutions function independently and do not integrate with response systems.

## 1.3 Research Gap

Few flood monitoring systems deal with post-disaster health management, despite the fact that many of them concentrate on flood prediction and early warning.

In large-scale disasters, current methods primarily rely on manual medical assessments, which are labour- intensive and slow. Disease outbreaks are more likely in many flood situations because medical personal is unable to promptly evaluate everyone's health.

Furthermore, while health risk monitoring is frequently handled independently, current disaster management platforms mainly concentrate on infrastructure monitoring, evacuation planning, and rescue coordination.

There is a clear research gap in integrating automated health risk detection with disaster response systems because most health monitoring systems only rely on self-reported symptoms, which may not always provide accurate information. Visual indicators like skin discoloration, facial swelling, fatigue, or eye redness can provide important early signs of illness but are rarely analyzed automatically.

The proposed system fills this gap by combining:

- Symptom-based health reporting
- Facial scan analysis
- Machine learning-based disease risk estimation
- Real-time dashboard visualization for medical team

This integrated approach enables early identification of flood-related health risks, allowing healthcare teams to respond quickly and prevent potential disease outbreak.

# 2. OBJECTIVE

## 2.1 Main Objective

To create and put into use an intelligent post-flood disease risk detection module that evaluates symptom reports and facial scan indicators in order to detect possible health risks and facilitate prompt medical intervention in communities affected by flooding.

## 2.2 Specific Objectives

- Create a digital data collection platform that uses a web or mobile interface to collect symptom-based health data from people living in flood-affected areas.
- Create a facial scan analysis system to detect outward signs of illness, such as fatigue, swelling, skin discoloration, and red eyes.
- Using collected symptom and facial scan data, develop a disease risk prediction model that uses machine learning and rule-based logic to estimate the degree of health risk.
- To help healthcare teams prioritize patients who need emergency care, generate automated health alerts and medical recommendation.
- Create a centralized monitoring dashboard to help healthcare teams monitor possible disease threats following flood events and to visualize health risk levels.
- Assess how well the suggested module support early post-flood disease risk detection and enhances disaster healthcare response.

# 3. METHODOLOGY

## 3.1 Research approach

The study undertakes a DSR method that emphasizes the designing, development, and evaluation of a unique technology that helps solve a real-world problem. The proposed module seeks to assist in the early detection of post-flood disease risks by incorporating symptoms reporting and facial scan analysis in a disaster management scenario.

The reason for choosing the design science method for this study is that it involves the development of a product or a module, along with evaluating the same by means of experimental testing or performance analysis, The research method for this study involves identifying the problem, designing the system, implementing the machine learning algorithms, and then evaluating the module that has been developed.

## 3.2 Technical methods and algorithms

The system proposed is based on the analysis of symptoms related to health conditions and facial scan analysis to determine the possible risks of disease for people affected by floods.

The first step is to collect data related to symptoms related to health conditions using a digital interface. The symptoms related to health conditions may include fever, skin irritation, tiredness, diarrhea, swelling, etc.

The second step is to collect facial scan images related to people affected by flood using a digital interface. The images will be processed to determine the possible signs related to health conditions such as skin irritation, facial swelling, eye irritation, tiredness, etc.

The data collected for symptoms related to health conditions and facial scan images will be processed using a disease risk prediction model. The possible disease risk prediction model is based on rule-based decision tables or machine learning algorithms like Random Forest or SVM.

The possible results obtained using the disease risk prediction model can be used to generate health alerts for people affected by flood.

## 3.3 Tools and platforms

Several software tools and platforms will be used to implement the proposed module.

- Python will be used as the main programming language dur to it's strong support for machine learning and data analysis.
- Scikit-learn, TensorFlow, or PyTorch will be used to develop and train the machine learning models.
- OpenCV will be used for facial image processing and visual feature detection.
- Pandas and NumPy will support data preprocessing and analysis.
- React will be used to develop the user interface for symptom data entry and dashboard visualization.
- Node.js will be used to handle backend services and data communication.
- MYSQL database will be used for storing symptom data, facial scan results, and prediction outputs.

All these technologies will offer extensive support for developing an AI module, as well as for data handling and system integration within a web-based system.

### 3.4 Data collection and ethical considerations

The data for the proposed system will be collected from individuals in the flood-affected area using the system's interface. The data that the proposed system collects may include the individual symptoms and their facial scan images for detecting health indicators.

In order for the proposed system to be ethical, it will follow these ethical guidelines:

- The proposed system will ensure that user contest id obtained for the collection of their symptoms or facial scan data.
- The proposed system will anonymize the personal data collected from individuals in order to protect their privacy.
- The proposed system will only collect necessary information form individuals for disease risk estimation.
- Dara will be stored securely and accessed only for research and disaster response purposes.

These measures help ensure that the system follows responsible data handling and privacy protection practices.

### 3.5 Validation strategy and performance metrics

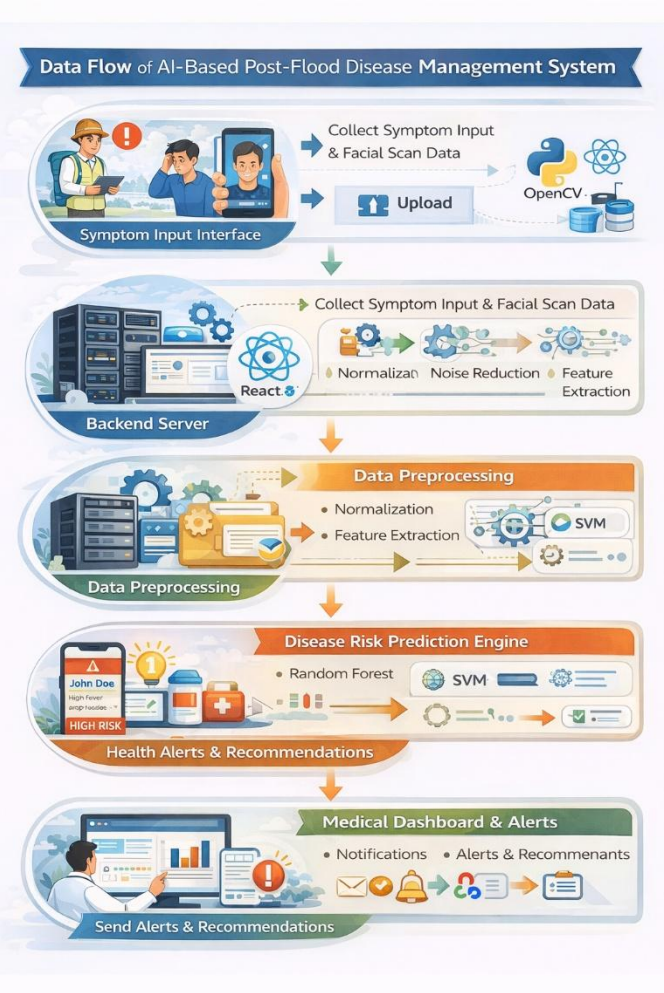
The performance of the proposed disease risk prediction module would be validated based on the result obtained from the experimental validation process. The dataset would be split into training data and test data to determine the predictive ability of the system.

The following metrics would be used to evaluate the performance of the system based on the common metrics used for machine learning classification:

- Accuracy – The accuracy of the system would be measured based on the correctness of the predictions made.
- Precision – The precision of the system would be used to determine the number of predictions made for high-risk cases.
- Recall (Sensitive) – The ability of the system to identify people who are actually at risk would be measured.
- F1-Score – The F1-score would be used to determine the performance of the system based on precision and recall.

The performance of the system would also be used to determine the usability of the system to ensure that it is capable of operating in real time.

*Figure 1 - Process of system*



# 4. HIGH-LEVEL SYSTEM ARCHITECTURE

## 4.1 Overview of proposed solution and its components

The proposed research is intended for an intelligent post-flood disaster response and recovery management system. The proposed solution is intended for various stages of disaster response, integrating various functional components of flood monitoring, rescue operations, relief delivery, and post-flood health risk management. These components are integrated into a single system that offers a coordinated response to flood-affected communities.

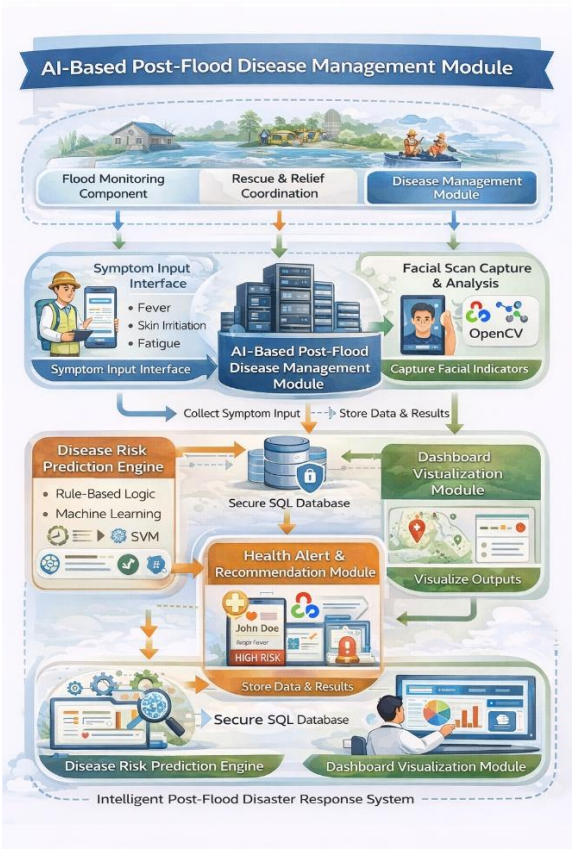
The proposed individual components of the system is the Post-Flood Disease Management Module. This component is designed to identify the possible disease risks of the affected persons. This module will gather information on the symptoms of the patients and the facial scan data. The information gathered with undergo a rule-based system and machine learning approach to generate health alerts and recommendations to the medical team. The information will be shown on a centralized dashboard.

The major parts of the proposed component include:

- Symptom Data Collection Interface
- Facial Scan Capture and Analysis Unit
- Disease Risk Prediction Engine
- Health Alert and Recommendation Module
- Dashboard Visualization Module
- Databased and Data Management Layer

## 4.2 System diagram of components

*Figure 2 - System Diagram*




4.3 Highlight the individual component and its parts

The main focus of this research is the Disease Management Module. The main focus of this module is that it is the healthcare support component in the overall disaster management system. The main difference between this module and others is that this module specifically deals with post-flood disease risk identification. The component comprises the following sub-parts:

- Symptom Input Interface
  This part allows individuals who have been affected by the disaster or officers in the field to input health symptoms via a web or mobile interface. The interface is used for collecting information such as fever, diarrhea, cough, skin irritation, swelling, tiredness, and many other health complaints.
- Facial Scan Capture and Analysis
  This unit captures a face image using a camera-enabled device. It also analyzes the visible symptoms such as skin discoloration, swelling, eye redness, or fatigue-related symptoms. Image processing techniques are used to identify the features relevant to health assessment.
- Disease Risk Prediction Engine
  This is the core part of the analysis for the module. In this part, the symptom data and the extracted facial indicators are used to determine the disease risk level for each individual. This analysis can be carried out using a rule-based logic and a machine learning model for classification into different levels like low, medium, or high.
- Alert and Recommendation Module

Based on the predicted risk level, the system generates alerts and basic recommendations for healthcare responders. This helps medical teams identify cases that require urgent attention.

- Dashboard Visualization Module
  This component offers a summary of health risks in a chart, indicators, and specific cases in a centralized dashboard. It allows medical teams and administrators to monitor the health status of the affected population more efficiently.
- Database and Data Storage Layer
  All symptom records, facial scan results, prediction outputs, and alert information are securely stored in the system database for future access, monitoring, and analysis.

### 4.4 Component interactions

The proposed component will interact with the users as well as the entire disaster response system. The users or the officers will input the required data regarding the health of the people affected by the disaster. The data will be inputted to the system through the symptom input interface and the facial scan capture interface. The input data will be sent to the disease risk prediction engine for analysis.

The prediction engine will process the input data and send the results to the alert generation module and the dashboard visualization module. The healthcare staff will use the dashboard to view the determined risk level. The staff will then determine the people who need urgent attention based on the determined level. In the context of the entire system, the proposed component will also be able to seek support from other modules. For example, the level of flooding or the area affected by the flooding from the flood monitoring component will help the healthcare staff determine the areas that are more vulnerable. The rescue and relief coordination component will also use the results from the proposed component to determine the people who are more vulnerable.

### 4.5 Data flow explanation

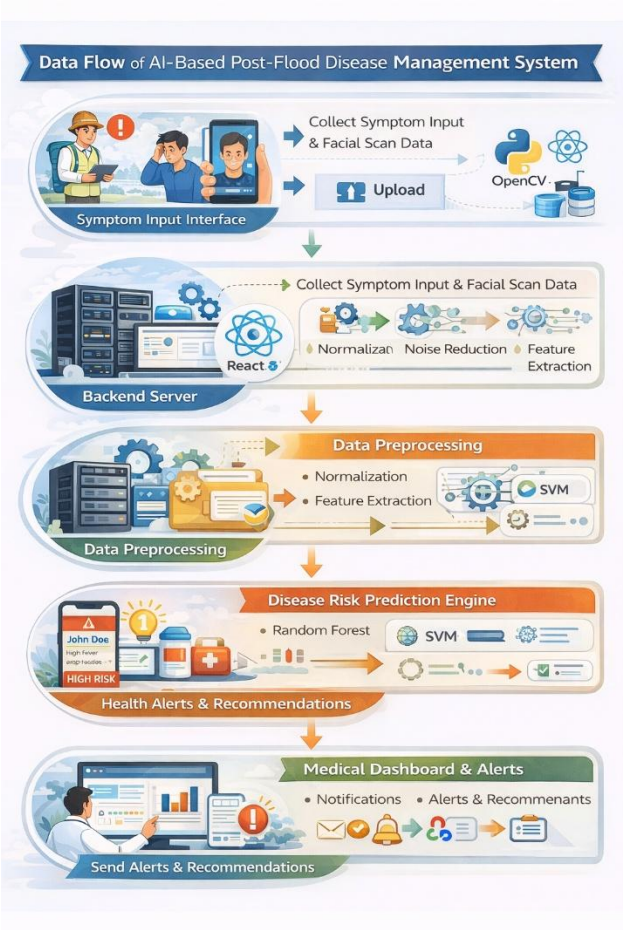
The data flow of the proposed system can be initiated when a user enters their symptom data and facial scan data via the web or mobile platform. The data is then sent to the backend server for preprocessing. The symptom data is preprocessed to format the data properly, while the facial scan data is processed to extract the appropriate visual indicators.

The data is then sent to the disease risk prediction module. The module processes the data using appropriate rules to determine the risk level of the patient. The results are then stored in the database and sent to the alert and recommendation module.

The results are then presented to the user via the dashboard in an appropriate format for the healthcare staff to understand. The results are then used to monitor the patient based on the data presented to the staff.

The data flow of the proposed system can be explained as follows:

*Figure 3 - Data Flow*




4.6 Scalability and security considerations

The architecture that is proposed is suitable for scalability to ensure that the system can accommodate a large number of users and health records in the case of a large-scale flood situation. This is due to the modularity of the architecture that allows each component to be improved or expanded. For example, the facial analysis component can be improved in the future with the implementation of more advanced image processing models. The dashboard component can also be expanded to accommodate the monitoring of various flood situations in different regions simultaneously.

In terms of security considerations, the proposed system is dealing with sensitive health-related information. Therefore, the privacy and protection of the data is important. The data that is gathered from the users should be done with the consent of the users. The personal data that is gathered should be minimized. The dashboard should be accessible to authorized persons only. Secure communication should also be implemented to ensure that the data is not misused. The proposed module is practical and suitable for the disaster response scenario.

# 5. USER REQUIREMENTS

## 5.1 Stakeholder identification

The proposed post-flood disease management module has a number of key stakeholders who interact with the system in different ways. The importance of these stakeholders lies in the fact that the system is intended to serve both the affected individuals and the healthcare responders in a post-flood disaster scenario.

Affected Individuals are the primary data providers for the system. Affected individuals can use the web or mobile interface to input the symptoms and facial scan data associated with their health condition in a post-flood disaster scenario.

Medical Teams and Healthcare Officers are the primary end users for the proposed module. Medical teams and healthcare officers use the system's dashboard to monitor health risks and take appropriate medical action.

The Disaster Management Authorities utilize the system to get a general idea about the health-related risks present in the affected communities. This helps them coordinate the disaster response activities effectively.

The System Administrators are responsible for maintaining the system, controlling user access, monitoring data storage, and ensuring the safe usage of the system.

The Researchers and Developers may utilize the data collected by the system for improving the model, evaluating the system, and further development of intelligent disaster healthcare solutions.

## 5.2 Functional requirements

The functional requirements describe the main services and tasks that the system must perform.

FR1. User Registration and Authentication

Authorized users such as healthcare teams and administrators to securely log in to the platform.

FR2. Symptom Data Submission

Allow affected individuals or field officers to enter symptom-based health information through a web or mobile interface.

FR3. Facial Scan Capture

Allow users to upload or capture facial images for visual health indicator analysis.

FR4. Symptom Data Storage

Store submitted symptom records securely in the database for further analysis and monitoring.

FR5. Facial Image Processing

The system shall process facial scan images to detect visible indicators such as swelling, akin discoloration, eye redness, and fatigue signs.

FR6. Disease Risk Prediction

Allow analyze symptom data and facial scan results using rule-based logic and machine learning techniques to estimate the individual's disease risk level.

FR7. Risk Classification

Classify health risk into categories such as low, medium, and high risk.

FR8. Alert Generation

Generate health alerts for cases identified as high risk or requiring urgent medical attention.

FR9. Recommendation Generation

Provides basic recommendations or follow-up guidance for healthcare teams based in the predicted risk level.

FR10. Dashboard Visualization

Display disease risk indicators. Alerts, and case summaries on a centralized dashboard for medical teams.

FR11. Search and Filter Records

Allow authorized users to search, filter, and review health records based on location, risk level, or reported symptoms.

FR12. Report Management

The system shall allow healthcare teams or administrators to access summarized reports of diseases risk pattern in flood-affected communities.

FR13. User Role Management

Support role-based access control for different user types such as admin, medical officer, and field officer.

FR14. Secure Data Access

Ensure that only authorized users can access sensitive health records and prediction outputs.

## 5.3 Non-functional requirements

The non-functional requirements identify the quality attributes that the system needs to ensure.

### 5.3.1 Security Requirements

The system shall ensure the safety of sensitive health data by means of secure authentication mechanisms. Personal data shall be safely stored, and unauthorized access shall be restricted. Data transmission between the user interface and backend shall also be ensured.

### 5.3.2 Performance Requirements

The system shall ensure that symptom data input and facial scan data input are processed within an acceptable time frame to enable near-real-time health monitoring. The response time for dashboard updates and alert generation shall be sufficient to enable disaster response activities.

### 5.3.3 Usability Requirements

The user interface shall be simple, clear, and easy to understand for both the affected individuals and the healthcare teams. The system shall ensure that the user interface is not complex for users who are not technology-savvy.

### 5.3.4 Reliability Requirements

The system shall ensure that it is reliable and accurate for accessing the records that have been submitted, the predictions made, and the alerts generated during disaster response scenarios.

### 5.3.5 Scalability Requirements

The system must be able to manage a growing number of users, submitted symptoms, and facial scan records during major flood incidents without compromising performance.

### 5.3.6 Availability Requirements

The system must be available when needed by the healthcare team or disaster response team during emergency operations, especially during emergency operations when decision-making is critical.

### 5.3.7 Maintainability Requirements

The system must be designed in a way that it is easy to maintain in the future by allowing for updates or improvements on the model.

### 5.3.8 Privacy Requirements

The system must be able to collect only relevant data for disease risk analysis while ensuring the privacy of users by anonymizing the data or allowing access only to a limited number of identifiable data.

# 6. COMMERCIALIZATION PLAN

- Target market and customer persona
  The target market for this system would be disaster management organizations in government agencies, healthcare centers, and other agencies involved in disaster response management. Such organizations would include the DMC in Sri Lanka, Ministry of Health, local government agencies, and emergency response teams. The system would be used to monitor health risks after a flood disaster.

  Secondly, NGOs, international agencies, and other agencies involved in disaster response management would also benefit from this system. Such agencies would require efficient tools to identify people in danger and provide timely medical assistance.

  The main user personas include:
  1. Disaster management officers who monitor affected regions and coordinate response activities.
  2. Healthcare professionals and medical teams who need to identify individuals at high risk of disease.
  3. Fiels officers and relief workers who collect symptom data from affected communities.
  4. Public health administrators who analyze health risk patterns during disaster recovery.

- Value proposition
  The proposed system provides a smart system of health monitoring for post-flood disaster situations. While the traditional systems of disaster management are designed to handle rescue operations and infrastructure monitoring, the proposed system provides a module for the early detection of diseases.

  The value of the proposed system includes:
  1. Early detection of the risk of diseases occurring as a result of the flood.
  2. Early response to medical emergencies and the prioritization of individuals at a greater risk of disease.
  3. The use of symptom reporting and facial scan analysis for the detection of health risks.

  Real-time monitoring of the health of individuals affected by the flood, as well as the coordination of the response of the disaster management teams.

- Revenue model
  The proposed system has the following options for commercialization: One of the options for the commercialization of the proposed system is the SaaS model, whereby the government or the disaster management organizations would subscribe to the system and would have to pay a periodic fee for the use of the system.

  Another model for the commercialization of the proposed system would be the government or institutional license model, whereby the government or the disaster management organizations would purchase the system for use.

  The proposed system could also be commercialized through partnerships with international organizations, whereby the system would be implemented by the humanitarian aid organizations

or the global health organizations, and they would sponsor the system for use in the disaster-prone areas.

- Cost estimation
  The estimated costs that are involved in developing and deploying the proposed system include hardware resources, cloud services, and development costs. Hardware resources include the costs of hardware that are used in developing the system and in testing the facial scan. Cloud services are required in hosting the application, storing data, and enabling real-time access to the dashboard.

  Development costs include costs related to developing software, training the machine learning model, system testing, and deployment. Although the initial development of the system is academic-based, the commercial version of the system could be subject to additional costs in enabling system scalability, security, and maintenance.

- Pricing strategy
  The pricing model for the proposed system can be based on the scale of deployment and the number of users.

  A subscription model can be adopted wherein a fee will be charged for access to the system. Large government organizations can be provided with an enterprise licensing model, which will allow them to access the system as many times as they want for their disaster management activities.

  For developing countries or humanitarian organizations, a discounted pricing model can be adopted.

- Competitive advantage
  The proposed system has various competitive advantages compared to the existing disaster monitoring systems.

  The first advantage is the integration of health risk monitoring in disaster management systems, which is not considered in the existing disaster monitoring systems. The second advantage is the reliability of the proposed system in detecting risks of diseases using the facial scan analysis compared to the existing disaster monitoring systems, which only rely on the symptoms of the patients.

  The proposed disaster monitoring system is beneficial in the sense that it provides real-time monitoring and enables the healthcare team to respond quickly in disaster situations. The integration of machine learning in the proposed system allows the prediction accuracy of the system to improve over time.

- Intellectual property considerations
  The proposed solution may include intellectual properties related to the software architecture, machine learning algorithms, and data processing techniques. These properties may be protected under software copyrights or patents based on the novelty of the implementation.

The source code and algorithms used by the system should be protected under software licensing agreements to avoid any unapproved use or modification. Additionally, the data collected by the system should be compliant with data protection and privacy laws to ensure ethical use of personal health information.

The protection of intellectual properties will ensure the sustainable development of the system while protecting the interests of the creators and other organizations involved.

# 7. BUDGET AND JUSTIFICATION

- Hardware costs
  Hardware resources are essential for the development process, testing, and data gathering. The development machine or computer is essential for the development of machine learning models and the web-based platform. A smartphone with a camera is essential for the facial scan data and the development of the smartphone interface. Finally, an external storage drive is essential for the safe storage of data sets, facial images, and backups during development.

- Software licenses
  It is worth noting that most of the software tools used in the proposed system are open-source, which will reduce the cost of the project as a whole. Python will be used as a primary programming language, with TensorFlow, Scikit-learn, and OpenCV supporting machine learning model development and image processing, respectively. In addition, React and Node.js will be used for web interface development. Since these software tools are open-source, no licensing cost will be incurred.

- Cloud services
  The need for cloud services arises for hosting the system and for data storage. A cloud platform like AWS, Google Cloud, or Azure will be utilized for hosting the backend and interface of the proposed system. In addition, cloud storage services will be needed for storing the symptom records and images from facial scanning that are collected from the users. This will ensure that the system is always available during disaster response scenarios.

- Human effort cost
  Human Effort refers to the time and efforts taken for the research, development, and testing of the system. Human Effort includes activities such as design of the system, development of machine learning models, and testing/validation of the system. Though the study is part of an academic assignment, estimating a value for human efforts represents the actual costs associated with development activities.

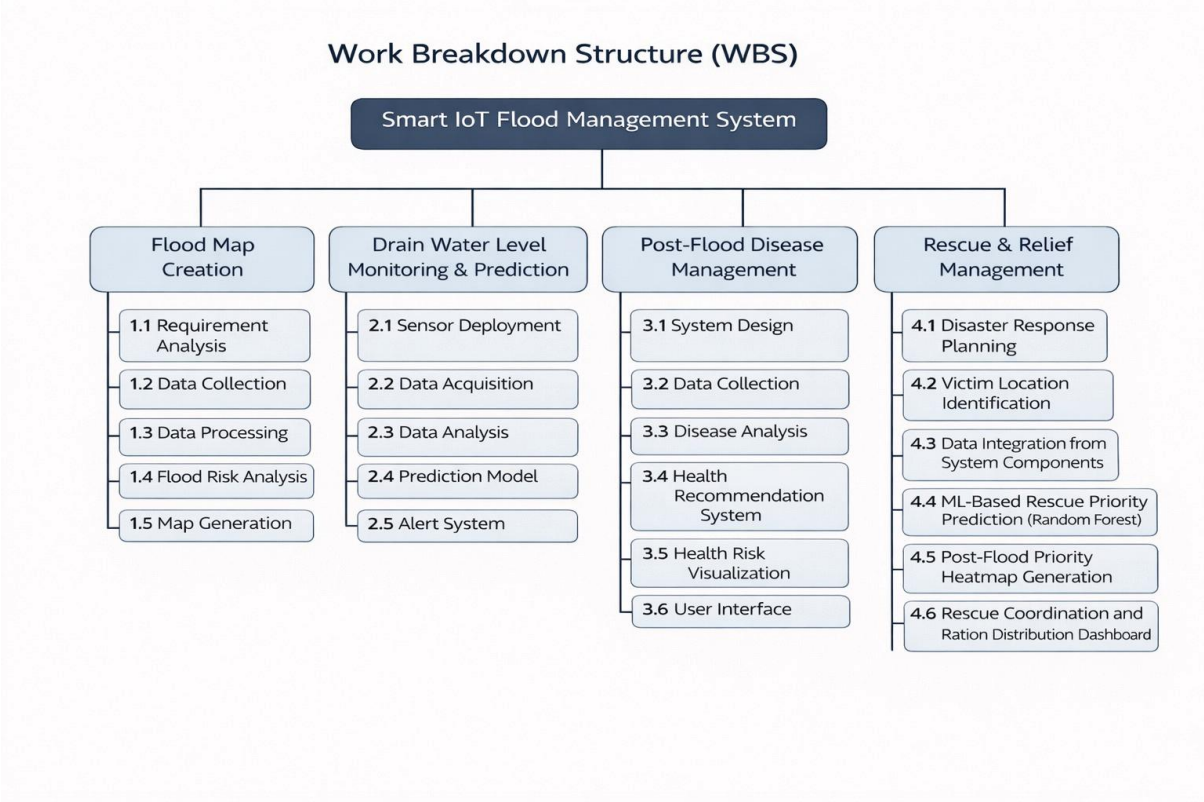
- Justification for each expense
  The proposed budget is intended to provide resources only for the essential resources needed to implement the proposed research project. The cost of the hardware tools is intended to support the system development and data collection. On the other hand, the software tools are mainly focused on using free technologies to reduce the cost. Additionally, cloud technology is included to provide the system the capability to perform real-time data processing. Finally, the cost of human efforts is the cost of the research needed to implement the proposed solution. The proposed budget provides adequate resources to implement the proposed solution successfully.

| **Category** | **Estimated Cost** |
|---|---|
| Hardware | Rs. 31130 |
| Software | Rs. 0 |
| Cloud Services | Rs. 62261 |
| Human Effort | Rs. 60000 |
| Miscellaneous | Rs. 31130 |
| Total Estimated Cost = | Rs. 183521 |

*Table 2 - Budget Table*

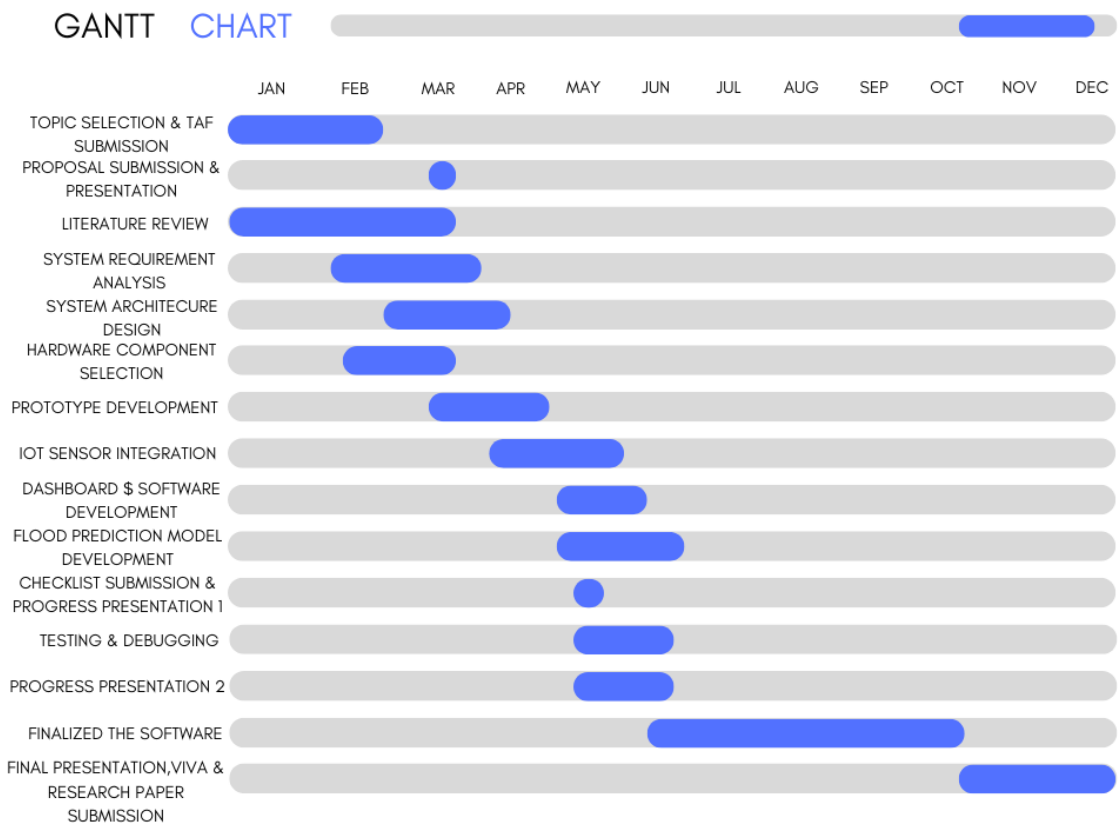
## 8. WORK BREAKDOWN STRUCTURE (WBS)

*Figure 4 - Work Breakdown Structure*



# 9. GANTT CHART

*Figure 5 - Gantt Chart*



## 10.REFERENCES LIST


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